

Energy dissipation in microtearing turbulence

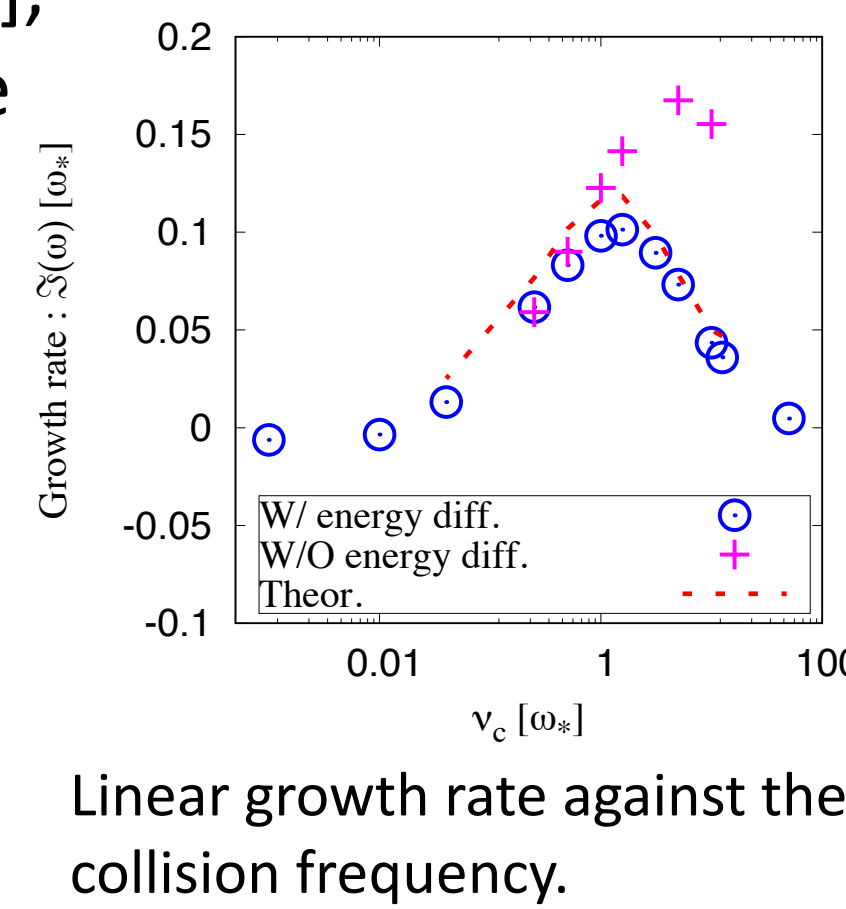
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Introduction

- Prediction of the electron anomalous thermal transport level due to microtearing (MT) turbulence driven by the electron temperature gradient is a crucial issue for designing future fusion devices.
- The thermal transport level for electrons is dependent on the saturation of the magnetic fluctuations. In the collisional regime with neglecting parallel dynamics, one expects $|\tilde{B}_x|/B_{z0} \sim \rho_e/L_{T0e}$, which is derived by a drift-kinetic model [1]. Gyrokinetic simulation work demonstrated that it follows the scaling of $|\tilde{B}_x|/B_{z0} \sim \rho_e/L_{T0e}$ in the semi-collisional regime [2].
- In previous study for the linear MT instability [3], we showed that MT is destabilized by the phase lag of the electromagnetic field and is survived even in a weakly collisional regime, $\nu_{ei}/\omega < 1$.
- In weakly collisional plasmas, phase mixing due to kinetic effects, such as finite Larmor radius (FLR) effects and Landau damping, creates a fine-scale structure in velocity space, and eventually leads to energy dissipation by the collision effect.
- We perform gyrokinetic simulations of MT turbulence using AstroGK [4] and investigate the saturation amplitude level of the magnetic fluctuations in the weakly collisional regime.



Question

Does linear or nonlinear phase mixing that is neglected in [1] affect to determine the saturation amplitude level of magnetic fluctuations in weakly collisional MT turbulence?

Quasi-linear theory

The saturation amplitude of the magnetic fluctuations is estimated by the balance between linear and nonlinear terms of the drift-kinetic equation using the quasi-linear analysis [1].

$$\omega = \omega_k + \frac{3}{4} i \Gamma \left(\frac{9}{2} \right) \eta_e \omega_k \omega_{*n} / \nu_{ei} - i (3\sqrt{\pi}/8) k_y^2 \nu_{ei} c^2 / \omega_{pe}^2 - \frac{5}{12} i \Gamma \left(\frac{1}{2} \right) \sum_{k'} \frac{\nu_e^2}{\nu_e} \frac{|\vec{k} \cdot \vec{B}_{k'}|^2}{B^2} \left(\frac{\omega_{*n} - \omega_{*n'}}{\omega_k + \omega_{k'}} \right) \eta_e \quad (1)$$

This relation is obtained by neglecting the electrostatic potential, parallel dynamics, and assuming the collisional regime $\nu_{ei}/\omega \gg 1$. the magnetic fluctuation amplitude scales as $|\tilde{B}_x|/B_{z0} \sim \rho_e/L_{T0e}$. This scaling is independent of the collision frequency. Note that linear and nonlinear phase mixing effects are not included in this analysis.

Simulation results

The initial setup of MT simulations in AstroGK follows [5] that is work for the normal tearing mode ($\nabla T_{0e} = \nabla n_{0e} = 0$), and the parameter setup follows [3].

Electron temperature and density gradients are, respectively, taken in the x direction as $L_{T0e}^{-1} = -\partial(\ln T_{0e})/\partial x$, $L_{n0e}^{-1} = -\partial(\ln n_{0e})/\partial x$, yielding the diamagnetic drift in the y direction with the frequency $\omega_{*T0e, n0e} = (k_y \rho_e / 2) (v_{th,e} / L_{n0e, T0e})$.

Measurement of the mag. fluc. amp.

We have measured the magnetic fluctuation amplitude $|\tilde{B}_x|/B_{z0} (L_{0e}/\rho_e)$ for the electron temperature gradient $\eta_e \equiv (\rho_e/L_{T0e})(L_{n0e}/\rho_e)$ in $\nu_{ei}/\omega_{*e} = 0.5, 1.56$. $\omega_{*e} \equiv \omega_{*T0e} + \omega_{*n0e}$

Summary

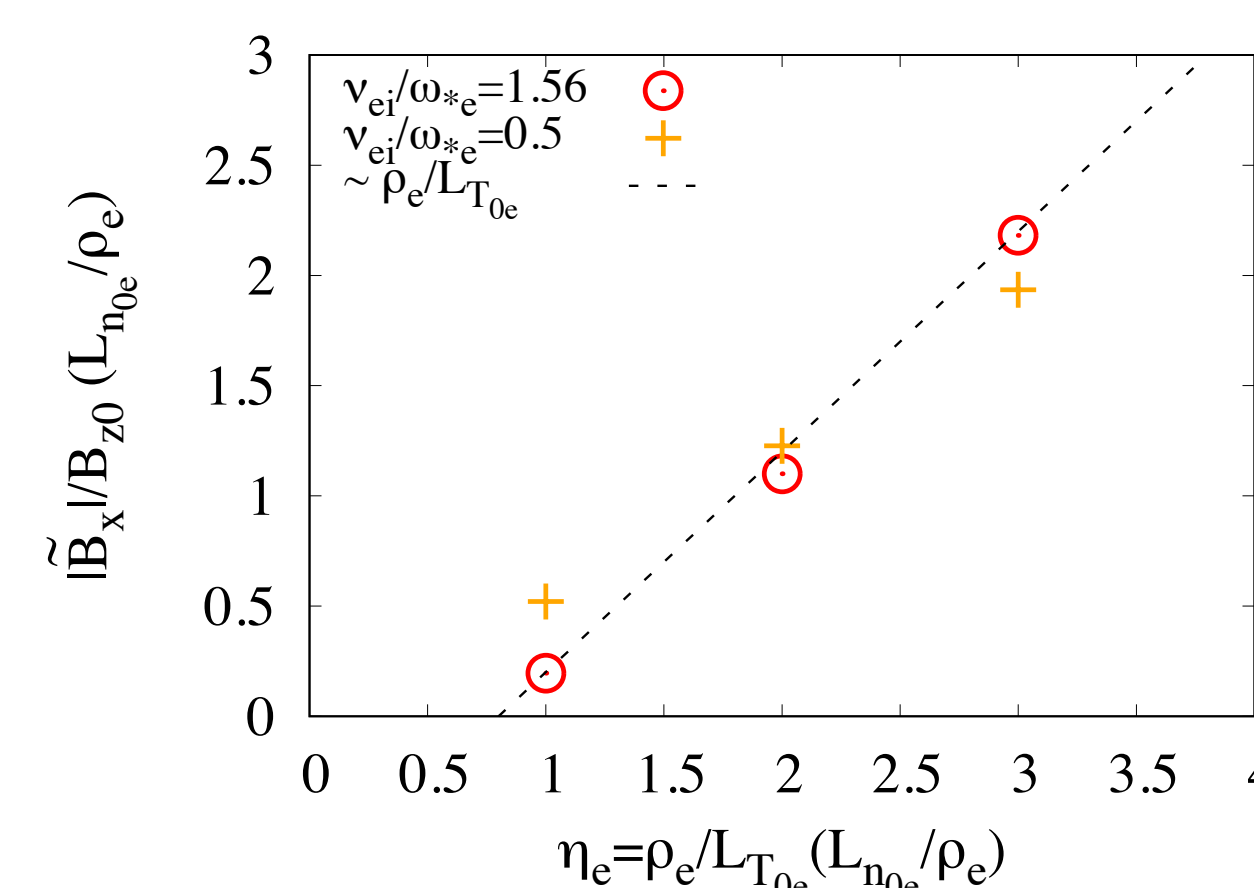
- We have performed MT turbulence by means of gyrokinetic simulations and investigated the saturation amplitude of the magnetic fluctuations in weakly collisional plasmas.
- In the semi-collisional regime, the magnetic fluctuation amplitude scales as $\sim \rho_e/L_{T0e}$. In the weakly collisional regime, it has been suggested that the saturation level of the magnetic fluctuations is larger than that of the semi-collisional case due to phase mixing. $\rightarrow$ If phase mixing contributes to the higher saturation level of the fluctuation, in weaker collisional plasmas, such as future devices, it may lead to stronger transport even if MT instability is weaker.
- More detailed analyses of the relationship between the magnetic fluctuation and phase mixing in MT turbulence are needed.

References

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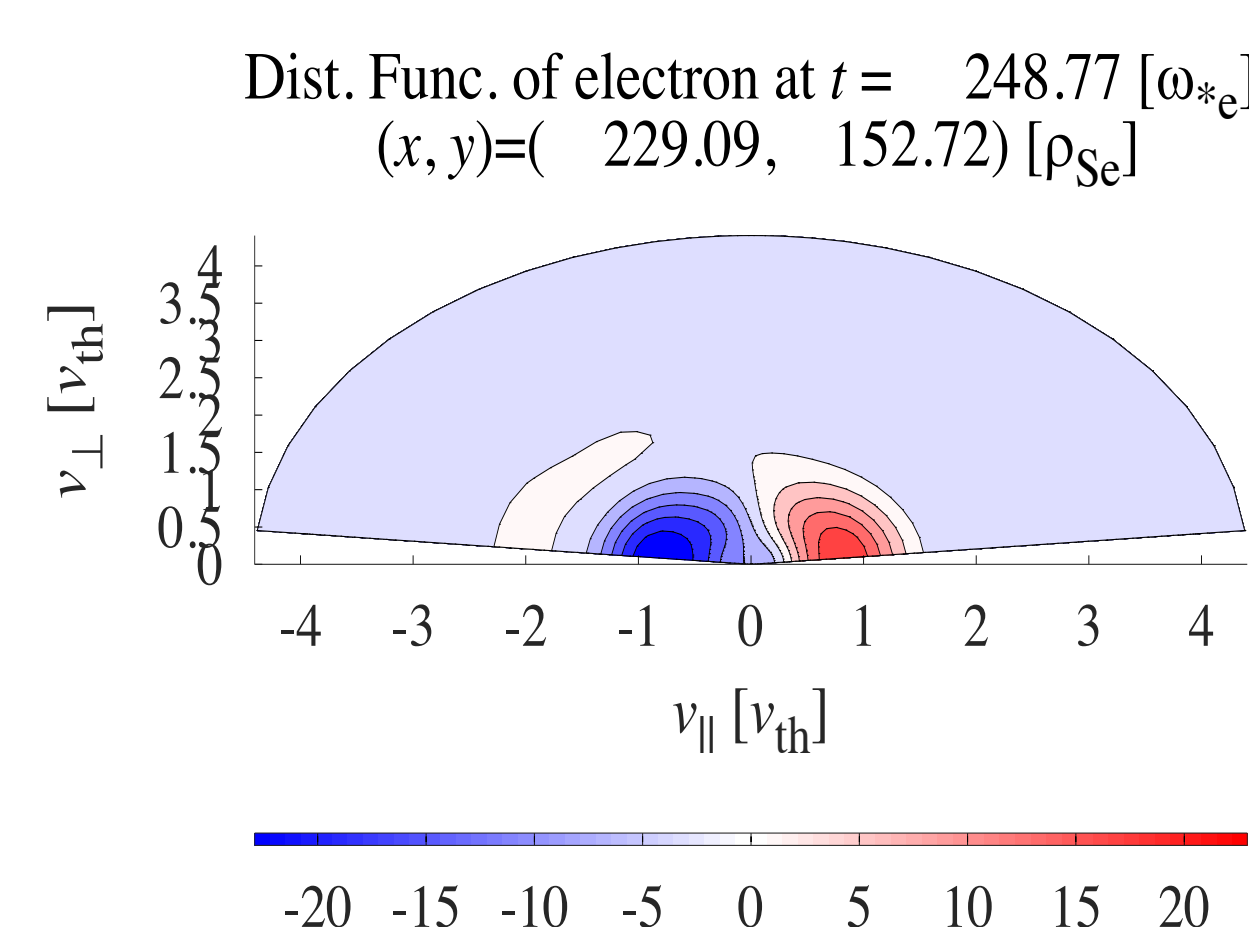
In the semi-collisional $\nu_{ei}/\omega_{*e} = 1.56$ case, the amplitude of magnetic fluctuations increases as the temperature gradient is stronger, and scales as $\sim \rho_e/L_{T0e}$ (Similar result was obtained in [2]).

Even in the weakly collisional $\nu_{ei}/\omega_{*e} = 0.5$ case, $|\tilde{B}_x|$ is proportional to η_e with a slope which is different from that of $\nu_{ei}/\omega_{*e} = 1.56$ case. However, the scaling of $|\tilde{B}_x|$ derived from (1) is independent of the collision frequency.

Phase mixing not considered in (1) may contribute to the saturation of $|\tilde{B}_x|$! Focusing on $\eta_e = 1$ case, $|\tilde{B}_x|$ of $\nu_{ei}/\omega_{*e} = 0.5$ is larger than that of $\nu_{ei}/\omega_{*e} = 1.56$.

If phase mixing leads to this result, in the weaker collisional regime, MT instability is suppressed but the saturation level of $|\tilde{B}_x|$ becomes higher due to the stronger phase mixing effect, and eventually transport may be promoted!

Velocity space structure



The structure of the electron distribution function in velocity space is shown in the left. $\eta_e = 1$, $\nu_{ei}/\omega_{*e} = 0.5$.

The distribution function is formed in the both $v_{\parallel}$ and $v_{\perp}$ directions. It suggests that linear and nonlinear phase mixing is occurring!

The detailed analyses of the structure in velocity space are required to quantitatively evaluate the phase mixing effect.